CRED Essential PYQ Practice Pack

Product Based / CRED

Study Hub Premium Placement Pack

This is original study material for preparation. It is not a copied official or proprietary company paper.

Premium Study System

- Track: CRED.
- Pack type: PYQ / PYQ Practice.
- Target outcome: selection-ready confidence for SDE / Software Engineer.
- Core preparation areas: DSA, CS fundamentals, OOP/LLD, HLD basics, behavioral stories, resume projects.
- Use this pack like a mini-book: concept, table, example, practice, analysis, revision.
- Every topic should produce proof: solved questions, mock score, mistake note, interview answer, or resume evidence.

Output Quality Rules

- Read the concept in your own words, then solve without looking at notes.
- Convert every important idea into a comparison table, checklist, or one-line rule.
- Mark important traps as Common Mistake before the mock, not after losing marks.
- Keep a PYQ-style log: question pattern, topic, time taken, mistake reason, retest date.
- Use simple English; add Hindi/Hinglish meaning in brackets when it helps recall.

Priority Matrix

Area	Premium Standard	Proof Before Moving On
DSA	Pattern, invariant, dry run, complexity	2 solved problems per pattern without hints
CS Core	OS, DBMS, CN, OOP with examples	Explain each topic in 60 seconds
Project	Problem, architecture, your contribution, tradeoff	2-minute story plus deep-dive answers
Communication	Crisp intro, structured HR stories, calm tone	5 recorded answers reviewed once

Exam Tip

- Interviewers reward clean reasoning more than memorized code. Say brute force, improve it, dry run, then discuss complexity.

Common Mistake

- Jumping into code without constraints creates wrong edge cases. Ask size, duplicates, sorted/unsorted, and expected output first.

Original PYQ Style Set

- This is original practice inspired by common placement patterns, not a copied company paper.

Original OA and Interview Practice

- This set is designed for CRED style product preparation: DSA depth, CS fundamentals, design thinking, project ownership, and behavioral clarity.

Section A: DSA Pattern Practice

- Q1. Return indices of two numbers adding to target. Explain hash map time and space complexity.
- Q2. Merge overlapping intervals and list the sorting invariant.
- Q3. Detect cycle in a linked list, then find the cycle start.
- Q4. Find shortest path in an unweighted graph and explain why BFS works.

- Q5. Find the longest substring without repeating characters. Explain sliding window movement.
- Q6. Search an element in a rotated sorted array. Explain the binary-search decision rule.
- Q7. Given daily temperatures, return days to wait for warmer temperature. Explain stack invariant.
- Q8. Count islands in a grid. Explain visited marking and boundary checks.

Section B: DP and Optimization

- Q9. Count ways to climb stairs with 1 or 2 steps, then optimize memory.
- Q10. Find maximum subarray sum and explain why local optimum works.
- Q11. Find coin change minimum coins. Define state, transition, base case, and impossible state.
- Q12. Given jobs with start, end, profit, select maximum profit non-overlapping jobs. Explain sorting + DP idea.

Section C: LLD and HLD

- Q13. LLD: Design a parking lot with classes, responsibilities, pricing, and edge cases.
- Q14. LLD: Design a rate limiter. Compare fixed window, sliding window, and token bucket.
- Q15. HLD: Design a tiny URL service. Cover APIs, storage, cache, collisions, and scaling.
- Q16. HLD: Design a notification system. Cover queues, retries, idempotency, and failure handling.

Section D: CS Core and Project Depth

- Q17. Explain database indexing and when an index can hurt performance.
- Q18. Explain process vs thread and how context switching affects performance.
- Q19. Explain HTTP status codes, cookies, JWT, and session tradeoffs.
- Q20. Describe a time you handled ambiguity while preparing for CRED. Use a real example.

Model Hints

- Q1 hint: hash map stores complement or seen values; time O(n), space O(n).
- Q3 hint: Floyd cycle detection gives meeting point; reset one pointer to head to find cycle start.
- Q8 hint: DFS/BFS from every unvisited land cell; each cell is visited once.
- Q14 hint: mention data model, request key, time bucket, concurrency, and cleanup.

Solution Checklist

- State brute force first, then optimized approach.
- Confirm input constraints before coding.
- Dry run one normal case and one edge case.
- End with complexity and production tradeoffs.
- For design rounds, define requirements before drawing components.

How To Review

- Mark every question as solved, guessed, skipped, or wrong.
- Rewrite the idea behind every wrong question in one sentence.
- Re-attempt wrong questions after 24 hours and again after 7 days.

PYQ-Style Attempt Protocol

- Step 1: identify topic and difficulty before solving.
- Step 2: write the first approach, even if it is brute force or long method.
- Step 3: optimize only after the basic idea is correct.
- Step 4: dry run one normal case and one edge case.
- Step 5: add answer, time, mistake type, and retest date to the log.

Pattern References

- Common OA patterns: hash map lookup, interval sorting, BFS/DFS, binary search on answer, DP state transition, stack/queue simulation.

Answer Review Table

Mark	Meaning	Next Action
Solved	correct under timer	revise after 7 days
Guessed	answer came without method	solve two similar questions
Wrong	concept or execution gap	write mistake rule and reattempt tomorrow
Skipped	time or confidence issue	learn shortcut or move to lower difficulty

Execution Dashboard

Metric	Target	Review Frequency
DSA Questions	10-14 per week	every Sunday
CS Core Notes	4 topics per week	twice a week
Mock Tests	1-2 per week	same day analysis
Interview Practice	5 answers per week	record and review

Weekly Review Ritual

- Pick the top 5 mistakes, rewrite the correct rule, and solve one similar question.
- Remove one weak resume line or prepare one stronger project answer.
- Decide the next week from evidence, not mood.

Final Self Audit

Score	Meaning	Action
0-40	reading stage	finish basics and easy questions
41-70	practice stage	increase timed mocks and mistake review
71-85	interview-ready	polish project, HR, and weak topics
86-100	selection-ready	maintain revision and avoid overloading new material

Source Note

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